S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

AI Risk Engine

From Financial Events to Portfolio Stress

THE PROBLEM

Risk information is scattered across company filings and external sources. Portfolio managers must understand both the event and its possible portfolio consequences.

THE SOLUTION

An explainable AI pipeline that converts financial disclosures into company-level risk signals and evaluates their hypothetical portfolio impact.



Chinmayi Upadhyay | Manipal Institute Of Technology, Manipal

The intelligence pipeline

PRIMARY DISCLOSURE EVIDENCE

1. **SEC EDGAR 8-K filings**
   Primary disclosure text

2. **FinBERT sentiment**
   Positive / neutral / negative financial sentiment

3. **Event classification**
   Event categories from text and matched keywords

4. **Impact scorer**
   Severity + sentiment + evidence-based confidence, bounded to a 1–10 score

EXTERNAL CONTEXT (SUPPLEMENTARY)

5. **GDELT Global Knowledge Graph**

- Article counts, tone, source URLs
- Metadata only, not full article text
- Matched from local records; not a continuous live news stream

6. **Combined risk:** company impact score plus a deliberately bounded external-context adjustment

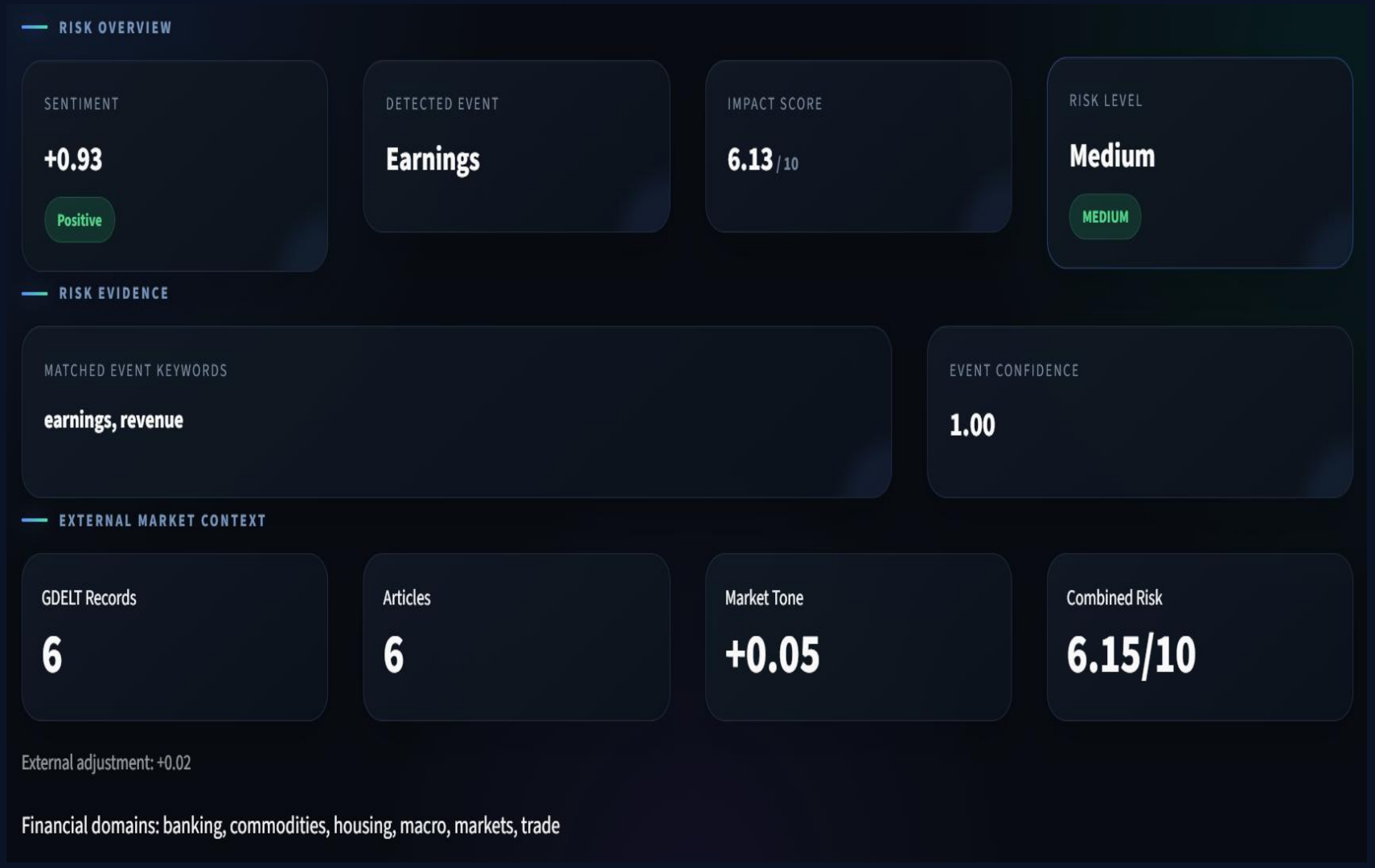
Architecture and implementation



IMPLEMENTED WITH

Python | PyTorch | Transformers / FinBERT | pandas | FastAPI | Streamlit

The working product



1. **Risk overview**
   combined score and risk band
2. **Evidence**
   filing text behind the signal
3. **GDELT context**
   records matched, tone, URLs
4. **Portfolio and scenarios**
   exposure and stress results

The working product



1. **Risk overview**
   combined score and risk band
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   filing text behind the signal
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   records matched, tone, URLs
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   exposure and stress results

Results: Apple Inc. illustrative prototype run

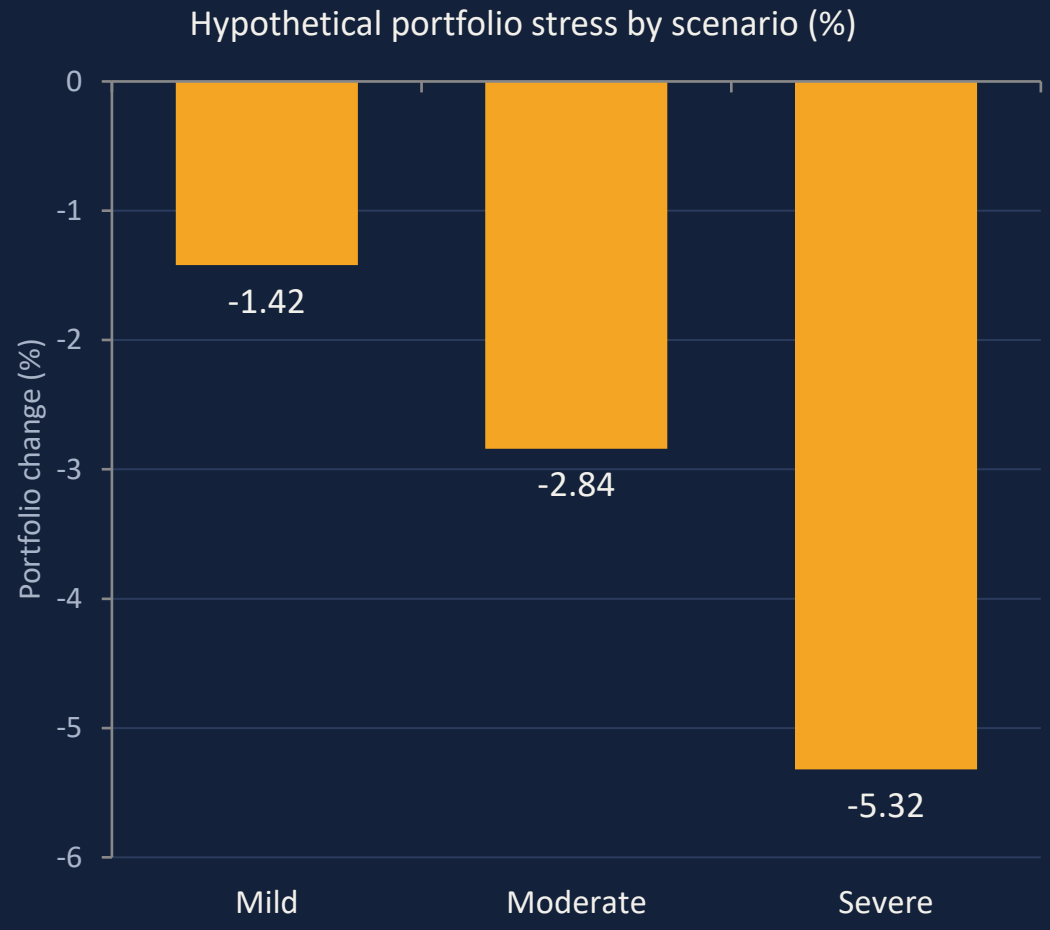
6.13 / 10	6	+0.05	≈ +0.02
SEC-derived impact score	GDELT records matched	Average GDELT tone	External-context adjustment

COMBINED IMPACT SCORE

6.15 / 10

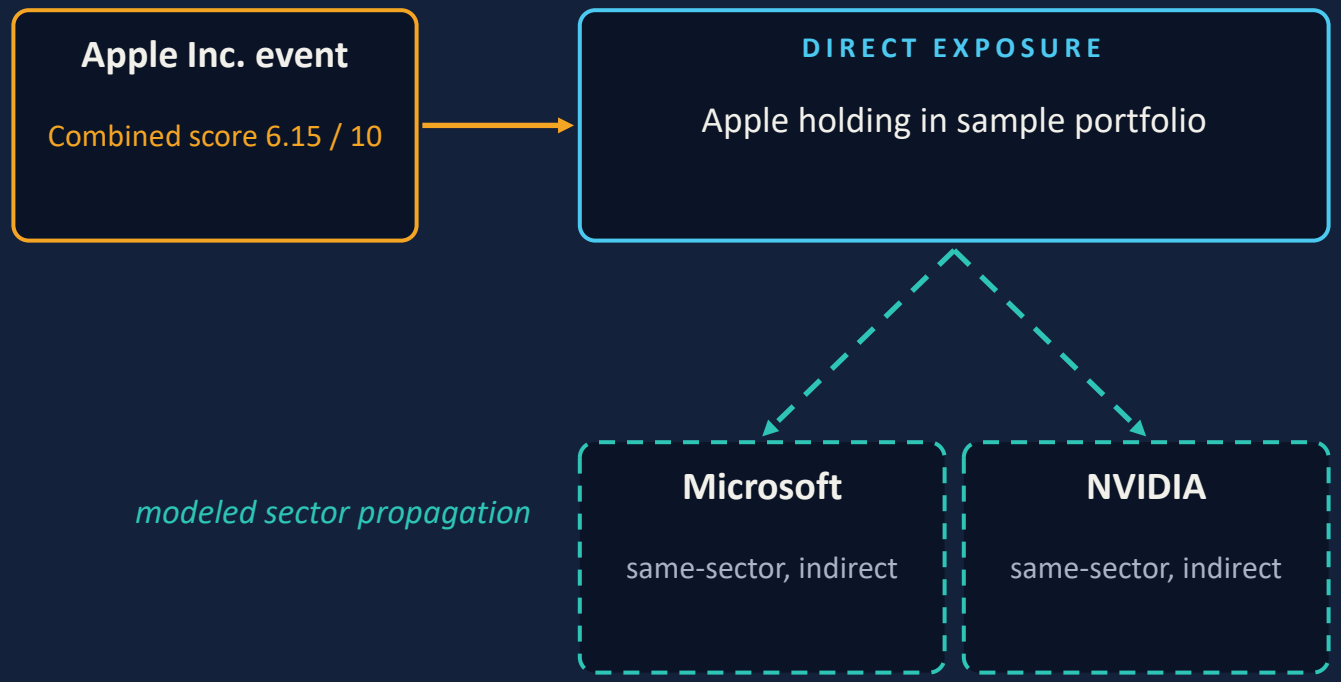
Band: Medium

Hypothetical scenario outputs under stated portfolio and propagation assumptions. Not forecasts, realized losses, or validated market predictions.



From company risk to portfolio consequences

ILLUSTRATIVE EXPOSURE PATH



Weights and propagation factors are assumptions in a sample portfolio.

WHY IT DIFFERS FROM SENTIMENT ALONE

1. **Traceable**
   each score links to filing evidence
2. **Bounded context**
   external adjustment is capped
3. **Inspectable**
   evidence visible in the dashboard
4. **Transparent**
   counterfactual, configurable shocks

Propagation factors and shock levels are model assumptions, not empirically calibrated causal relationships.

Impact, limitations, and demonstration

POTENTIAL USE CASES

1. **Analyst triage**
   prioritize filings by risk signal
2. **Event-driven portfolio review**
   see exposed holdings after an event
3. **Rapid hypothetical stress testing**
   compare mild, moderate and severe cases

LIMITATIONS

- Prototype, not a production trading or risk-management system
- GDELT context relies on available local records; not a live stream
- Portfolio weights and scenario shocks are illustrative assumptions
- Small manual evaluation sets give preliminary sanity checks, not broad validation

GITHUB Trackon24/AI-Risk-Engine

DEMO VIDEO https://youtu.be/g3wXb1_eD0M

PRESENTER Chinmayi Upadhyay, MIT Manipal

From disclosure to exposure: make the path from a financial event to portfolio risk inspectable.